

1. If $P(A \cup B) = P(A) + P(B)$, the events A and B are :
(a) Independent (b) Mutually exclusive
(c) Equally likely (d) Any arbitrary one
2. The Probability of getting number 5 in throwing a die is:
(a) 1 (b) $1/3$ (c) $1/6$ (d) $5/6$
3. A card drawn from a well shuffled pack of cards. The probability of getting a queen of club or king of heart is :
(a) $1/52$ (b) $1/26$ (c) $1/13$ (d) None
4. The probability of having atleast one head in 3 throws with a coin is
(a) $7/8$ (b) $3/8$ (c) $1/8$ (d) None
5. A coin is tossed and a die is rolled. The chance that the coin shows a head and the die shows 3 is :
(a) $1/8$ (b) $1/12$ (c) $1/2$ (d) 1
6. Five coins whose faces are marked 2,3 are thrown, the Probability of obtaining a total of 12 is :
(a) $5/16$ (b) $1/8$ (c) $3/16$ (d) None
7. The probability of getting 10 exactly in one throw with three dice is :
(a) $1/4$ (b) $3/8$ (c) $1/8$ (d) None
8. An experiment succeeds twice often as it fails. The chance that in the next six trials there will be at least four successes is :
(a) $240/729$ (b) $496/729$
(c) $233/729$ (d) None of these
9. A coin is tossed n times. What is the chance that the head will present itself odd number of times :
(a) $2^n/(n-1)$ (b) $2^n/n$
(c) $1/2$ (d) $1_n/2$
10. A and B throw a coin alternately till one of them sets a head and wins the game. What is the probability of winning the game by A :
(a) $1/3$ (b) $2/3$ (c) $1/6$ (d) $5/6$
11. If 8% of the bulbs produced by a factory are red, 2% red and defective. If one bulb is picked up at random then what is the probability that it is defective if it is already red :
(a) $1/8$ (b) $1/4$ (c) $3/4$ (d) None
12. In a college 30% students fail in physics, 25% fail in Maths, 10% fail in both. What is the probability that he fail in Maths or Physics.
(a) $1/4$ (b) $1/10$ (c) $2/5$ (d) $9/20$
13. If $P(A \cup B) = 0.9$, $P(B) = 0.4$. when A and B are independent events then $P(A) =$
(a) $1/3$ (b) $2/3$ (c) $5/6$ (d) $1/6$
14. If E_1 and E_2 are exclusive events then $P(E_1 \cap E_2) =$
(a) $P(E_1) + P(E_2)$ (b) $P(E_1) \times P(E_2)$
(c) 0 (d) None
15. If A and B are two events and
 $P(A) = 3/8$, $P(B) = 5/8$, $P(A \cup B) = 3/4$ then
 $P(A/B) =$
(a) $1/4$ (b) $2/5$ (c) $3/5$ (d) $1/5$
16. If E_1 and E_2 are exclusive events then $P(E_1/E_2) =$
(a) 1 (b) 0 (c) $1/2$ (d) None
17. An urn contains 6 balls of which 3 are white and 3 are black. They are drawn successively without replacement; what is the chance that colours are alternate :
(a) $1/20$ (b) $1/10$ (c) $1/5$ (d) $2/5$
18. One bag contains 5 white and 3 black balls and another bag contains 4 white and 5 red balls. Two balls are drawn from one of them, what is the chance that they are of different colours :
(a) $15/56$ (b) $5/18$
(c) $15/56 + 5/18$ (d) None of these
19. Two events A and B have probability .25 and .50 . The probability that both occur simultaneously is .14 . Then probability that neither A nor B occur is :
(a) .75 (b) .61 (c) .39 (d) None
20. A speak truth 60% cases and B in 70% cases. In what percentage they are likely to contradict each other in stating the same fact :
(a) 52% (b) 46% (c) 40% (d) 30%
21. A four figure number is formed from the digits 0,1,2,3,4,5, . Find the chance that number is divisible by 5 :
(a) $5/6$ (b) $2/5$
(c) $9/25$ (d) $3/20$
22. In solving any problem, odds against ' A ' are 4 to 3 and odds in favour of ' B ' in solving the same is 7 to 5 . Then probability that problem will be solved is :
(a) $5/21$ (b) $16/21$
(c) $15/84$ (d) $69/84$

- 23.** The chance of winning of two horses are $\frac{1}{3}$ and $\frac{1}{6}$ respectively. What is the chance that atleast one will win the same race :
(a) $\frac{1}{2}$ (b) $\frac{4}{9}$ (c) $\frac{5}{9}$ (d) $\frac{3}{4}$
- 24.** A and B appear for an interview for two vacancies for same post, Prob. of their selection are $\frac{1}{3}$ and $\frac{1}{5}$ respectively. Probability if any one is selected is :
(a) $\frac{8}{15}$ (b) $\frac{6}{15}$ (c) $\frac{14}{15}$ (d) $\frac{7}{15}$
- 25.** A problem in Maths given to 3 students A,B,C whose chances of solving are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ respectively. Probability that problem will be solved is :
(a) $\frac{1}{4}$ (b) $\frac{1}{24}$ (c) $\frac{23}{24}$ (d) $\frac{3}{4}$
- 26.** There are 4 addressed envelopes and 4 letters. Then the chance that all the letters are not mailed through proper envelop is :
(a) $\frac{1}{24}$ (b) 1 (c) $\frac{23}{24}$ (d) $\frac{9}{2}$
- 27.** If two dice are thrown find probability of getting an odd number on one and multiple of 3 on other is :
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{11}{36}$ (d) $\frac{13}{36}$
- 28.** In suffling a pack of cards 3 are accidentally dropped, then the chance that missing card should be of different suits :
(a) $\frac{169}{425}$ (b) $\frac{261}{425}$
(c) $\frac{104}{425}$ (d) None of these
- 29.** If one card is drawn then probability that the card is spade or a king :
(a) $\frac{16}{52}$ (b) $\frac{17}{52}$ (c) $\frac{20}{52}$ (d) None
- 30.** In a single throw of two dice what is the chance that sum is neither 7 nor 11 :
(a) $\frac{1}{2}$ (b) $\frac{107}{108}$ (c) $\frac{7}{9}$ (d) $\frac{2}{9}$

SCORE SHEET

a	b	c	d	a	b	c	d	a	b	c	d	a	b	c	d	a	b	c	d
1	☐	☐	☐	☐	7	☐	☐	☐	☐	13	☐	☐	☐	☐	19	☐	☐	☐	☐
2	☐	☐	☐	☐	8	☐	☐	☐	☐	14	☐	☐	☐	☐	20	☐	☐	☐	☐
3	☐	☐	☐	☐	9	☐	☐	☐	☐	15	☐	☐	☐	☐	21	☐	☐	☐	☐
4	☐	☐	☐	☐	10	☐	☐	☐	☐	16	☐	☐	☐	☐	22	☐	☐	☐	☐
5	☐	☐	☐	☐	11	☐	☐	☐	☐	17	☐	☐	☐	☐	23	☐	☐	☐	☐
6	☐	☐	☐	☐	12	☐	☐	☐	☐	18	☐	☐	☐	☐	24	☐	☐	☐	☐

PROBABILITY - 02

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| 1. (b) | 2. (c) | 3. (b) | 4. (a) | 5. (b) | 6. (a) | 7. (c) | 8. (d) | 9. (c) | 10. (b) |
| 11. (b) | 12. (d) | 13. (c) | 14. (c) | 15. (b) | 16. (b) | 17. (b) | 18. (c) | 19. (c) | 20. (b) |
| 21. (c) | 22. (b) | 23. (b) | 24. (d) | 25. (d) | 26. (c) | 27. (c) | 28. (a) | 29. (a) | 30. (c) |